$$\int x^3 \log x \; dx = \int u dv$$
$$= uv - \int v du$$
$$= \frac{x^4}{4}\log x - \frac{1}{4}\int x^3 dx$$
$$= \frac{x^4}{4}\log x - \frac{1}{4}\left(\frac{x^4}{4}\right)+c$$
$$= \frac{x^4}{4}\left[\log x - \frac{1}{4}\right]+c$$

Example 2.28

Evaluate $\int (\log x)^2 \, dx$

Solution:

Take $u = (\log x)^2$	and $dv = dx$
Differentiate	Integrate
$du = (2\log x)\left(\frac{1}{x}dx\right)$	$v = x$

$$\int (\log x)^2 \, dx = \int u dv$$
$$= uv - \int v du$$
$$= x(\log x)^2 - 2\int \log x \, dx \; \ldots (*)$$

For $\int \log x \, dx$ in (*)

Take $u = (\log x)$	and $dv = dx$
Differentiate	Integrate
$du = \frac{1}{x}dx$	$v = x$

$$= x(\log x)^2 - 2\int u dv$$
$$= x(\log x)^2 - 2\left[uv - \int v du\right]$$
$$= x(\log x)^2 - 2\left[x\log x - \int dx\right]$$
$$= x(\log x)^2 - 2x\log x + 2x + c$$
$$= x\left[(\log x)^2 - \log x^2 + 2\right]+c$$

Example 2.29

Evaluate $\int (x^2 - 2x + 5)e^{-x} dx$

Solution:

Successive derivatives	Repeated integrals
Take $u = x^2 - 2x + 5$	and $dv = e^{-x}dx$
	$v = -e^{-x}$
$u' = 2x - 2$	$v_1 = e^{-x}$
$u'' = 2$	$v_2 = -e^{-x}$

$$\int (x^2 - 2x + 5)e^{-x} dx$$
$$= \int u dv$$
$$= uv - u'v_1 + u''v_2 - u'''v_3 + \ldots$$
$$= (x^2 - 2x + 5)(-e^{-x}) - (2x - 2)e^{-x} + 2(-e^{-x}) + c$$
$$= e^{-x}(-x^2 - 5) + c$$

Exercise 2.5

Integrate the following with respect to x.

1. xe^{-x}
2. x^3e^{3x}
3. $\log x$
4. $x\log x$
5. $x^n \log x$
6. $x^5e^{x^2}$

2.1.5 Integration by substitution (or) change of variable method

Integrals of certain functions cannot be obtained directly, because they are not in any one of the standard forms as discussed above, but may be reduced to a standard form by suitable substitution. The method of evaluating an integral by reducing it to a standard form by suitable substitution is called integration by substitution.

Type: VI

1. $\int [f(x)]^n f'(x)dx = \frac{[f(x)]^{n+1}}{n+1} + c, n \neq -1$
2. $\int \frac{f'(x)}{f(x)}dx = \log|f(x)| + c$
3. $\int \frac{f'(x)}{\sqrt{f(x)}}dx = 2\sqrt{f(x)} + c$
4. $\int e^x [f(x) + f'(x)]dx = e^x f(x) + c$
5. $\int e^{ax} [a f(x) + f'(x)]dx = e^{ax} f(x) + c$

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